

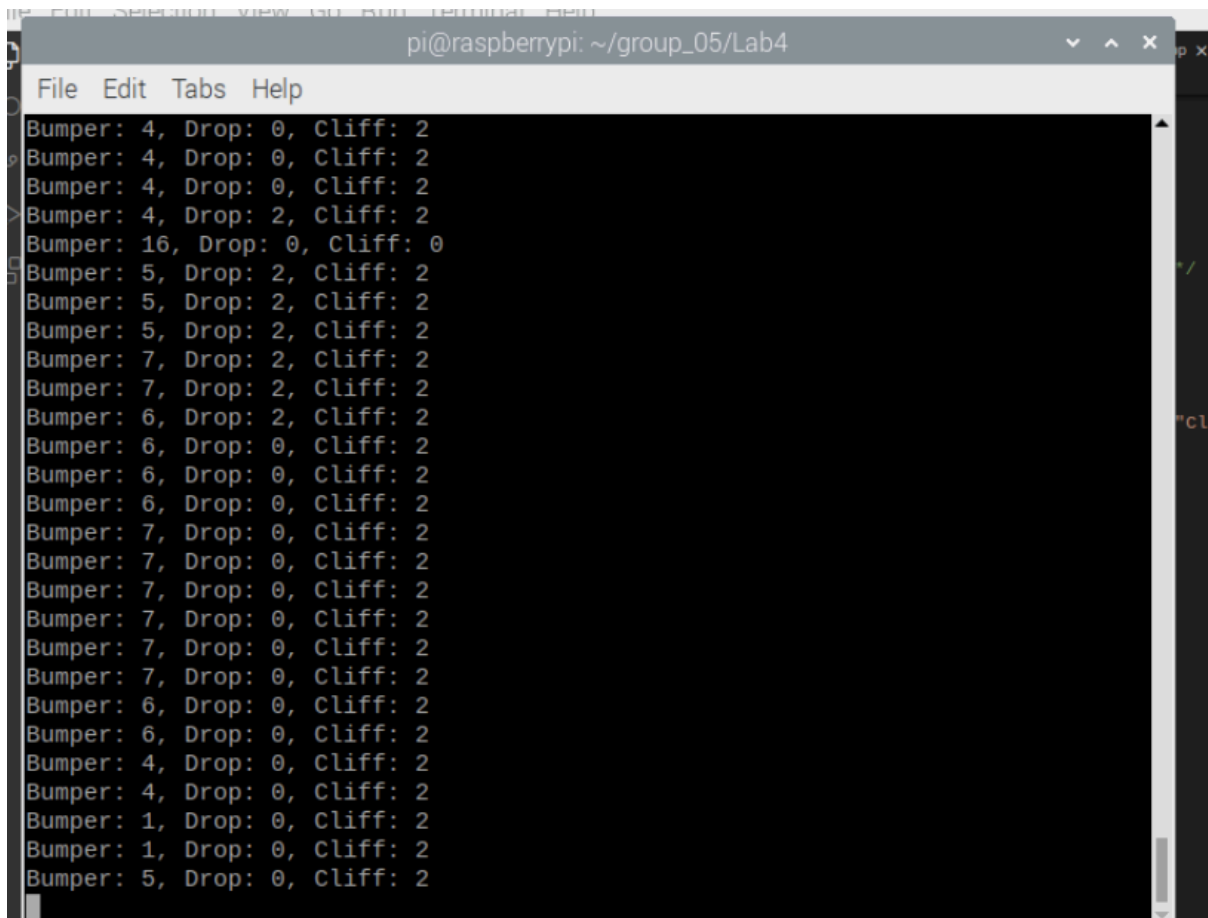
Lab 4: Autonomous Robot Motion Control with Sensors

Christian Landrum & Wyatt Probst

Exercise 1

1.1: Reading Kobuki Sensors

Exercise 1 was reading the Kobuki sensors using Serial communication and decoding what bumper or sensor was being triggered.

A terminal window titled 'pi@raspberrypi: ~/group_05/Lab4' displays a list of sensor readings. The window has a menu bar with 'File', 'Edit', 'Tabs', and 'Help'. The output consists of 25 lines, each representing a sensor state: 'Bumper: X, Drop: Y, Cliff: Z'. The values for X (Bumper) range from 1 to 16, Y (Drop) is either 0 or 2, and Z (Cliff) is either 0 or 2. The sequence of readings is: 4, 4, 4, 4, 16, 5, 5, 5, 7, 7, 6, 6, 6, 6, 7, 7, 7, 7, 7, 6, 6, 4, 4, 1, 1, 5.

```
pi@raspberrypi: ~/group_05/Lab4
File Edit Tabs Help
Bumper: 4, Drop: 0, Cliff: 2
Bumper: 4, Drop: 0, Cliff: 2
Bumper: 4, Drop: 0, Cliff: 2
Bumper: 4, Drop: 2, Cliff: 2
Bumper: 16, Drop: 0, Cliff: 0
Bumper: 5, Drop: 2, Cliff: 2
Bumper: 5, Drop: 2, Cliff: 2
Bumper: 5, Drop: 2, Cliff: 2
Bumper: 7, Drop: 2, Cliff: 2
Bumper: 7, Drop: 2, Cliff: 2
Bumper: 6, Drop: 2, Cliff: 2
Bumper: 6, Drop: 0, Cliff: 2
Bumper: 6, Drop: 0, Cliff: 2
Bumper: 6, Drop: 0, Cliff: 2
Bumper: 7, Drop: 0, Cliff: 2
Bumper: 7, Drop: 0, Cliff: 2
Bumper: 7, Drop: 0, Cliff: 2
Bumper: 7, Drop: 0, Cliff: 2
Bumper: 7, Drop: 0, Cliff: 2
Bumper: 7, Drop: 0, Cliff: 2
Bumper: 6, Drop: 0, Cliff: 2
Bumper: 6, Drop: 0, Cliff: 2
Bumper: 4, Drop: 0, Cliff: 2
Bumper: 4, Drop: 0, Cliff: 2
Bumper: 1, Drop: 0, Cliff: 2
Bumper: 1, Drop: 0, Cliff: 2
Bumper: 5, Drop: 0, Cliff: 2
```

Figure 1: Terminal Output of Kobuki Sensors

Exercise 2

2.1 Navigating from Point A to B Using Sonar

Exercise 2 was using Sonar to have the Kobuki travel through a path using the sonar sensors to avoid bumping into walls.

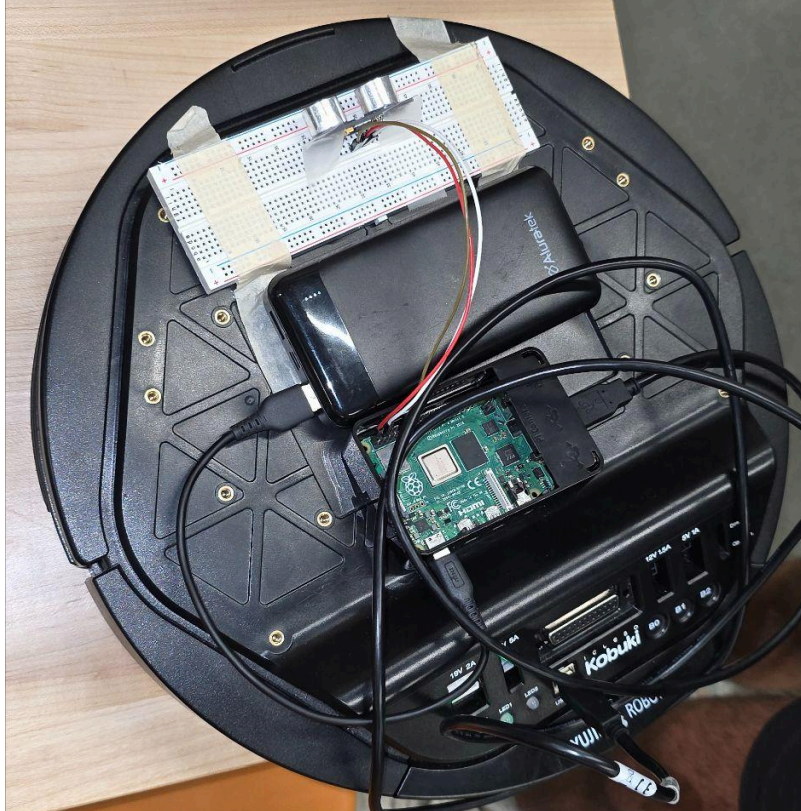


Figure 2: Sonar Connected to Kobuki

Exercise 3

3.1 Obstacle Avoidance and Random Motion

Exercise 3 was about making the Kobuki able to move in random directions and react to its environment using the sensors. We had to make it always move – avoiding long pauses or idle behavior. It must detect edges using cliff sensors and avoid falling off elevated surfaces. It should use bumper and sonar/cliff sensor data to detect obstacles and avoid getting stuck in corners or walls

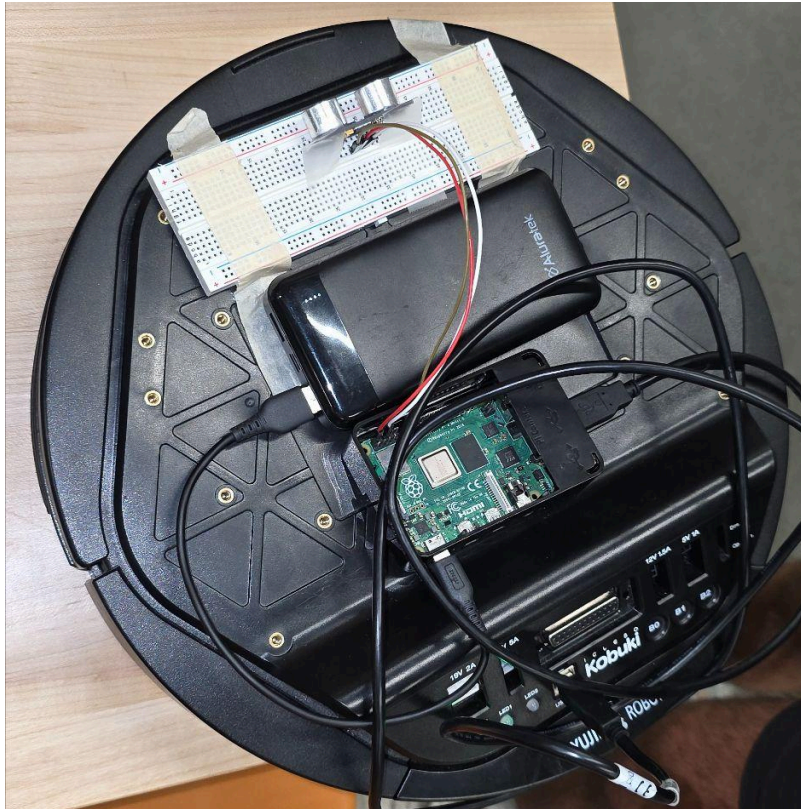


Figure 3: Kobuki Setup for Exercise 3

Supplemental Questions

Briefly summarize what you learned from this lab.

In this lab we learned about all the types of sensors on the kobuki robot, how to read them into our program, and how to interpret the values within our program to manipulate the robot to produce the desired output. We had to create our own logic with trial and error to allow the movement of the robot to be as efficient and smart as possible.

If you received out-of-range or missing values from the sonar sensor, what may be the reason for this?

Missing values from the sonar sensor can happen for a variety of reasons including maximum range exceeded, minimum range breached, sensor deadzones, sensor angle, or problems within the circuit.

Summarize the capabilities of sonar, bumper, wheel drop, and cliff sensors in robot navigation.

Sonar sensors are able to understand exactly how far a robot is from a surface and adjust accordingly. Bumper sensors can tell if a robot is currently running into something which is useful for blind spots that the sonar sensor may have missed or as a backup. The wheel drop sensors send information about whether or not the wheels are on the ground and can be used as a backup to the cliff sensors. The cliff sensors can tell if the robot is over an edge before it falls off and can send information to back up or turn before it falls. The combination of all of these sensors is often enough to allow the robot to roam freely and randomly without getting stuck.

ACKNOWLEDGMENTS

I certify that this report is my/our own work, based on my/our personal study and/or research and that I/we have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication. I/We also certify that this assignment/report has not previously been

submitted for assessment anywhere, except where specific permission has been granted from the coordinators involved.

Author-1 Signature **Christian Landrum**

Author-2 Signature **Wyatt Probst**

REFERENCES

- 1) Lab 4 Slides
- 2) Lab 4 Handout
- 3) Lab 4 Documents